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• ARTICLES •

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On the sum of two integral squares in the imaginary quadratic field $\mathbb{Q}(\sqrt{-2p})$

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Abstract We determine the sum of two integral squares over imaginary quadratic fields $\mathbb{Q}(\sqrt{-2p})$, where $p \equiv 1 \mod 8$ is a prime satisfying $2p = r^2 + s^2$ with $r, s \equiv \pm 3 \mod 8$.

Keywords integral points, ring class field, reciprocity law

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0 Introduction

The sum of squares over $\mathbb{Z}$ has been studied starting from Fermat, Gauss and Lagrange. All these classical results satisfy the local-global principle. Siegel [13] investigated the sum of integral squares over number fields by using circle method and obtained that the local-global principle still holds for the sum of more than three integral squares if the number fields are not totally real. Indeed, the local-global principle is no longer true for the sum of two or three integral squares over number fields. In [1] (see also [5]), the sum of three integral squares over all imaginary quadratic fields has been determined completely by using the Brauer-Manin obstruction. However, similar question will be more difficult for two integral squares over quadratic imaginary fields.

Niven [10] determined the sum of two integral squares over $\mathbb{Q}(\sqrt{-1})$. Nagell further studied the problem for $F = \mathbb{Q}(\sqrt{d})$, where d is one of the following twenty integers:

$$\pm 2,\ \pm 3,\ \pm 5,\ \pm 7,\ \pm 11,\ \pm 13,\ \pm 19,\ \pm 43,\ \pm 67,\ \pm 163$$

in [7] and [8]. His method follows Gauss' original ideal and essentially depends on the fact that the class number of $\mathbb{Q}(\sqrt{d}, \sqrt{-d})$ is 1 when d is one of the above integers. Obviously this method cannot apply for general quadratic fields.

Harari's result [4] ensures that the Brauer-Manin obstruction is the only one for such Diophantine equation. Xu and Wei [17,18] further constructed the idelic class group of $\mathbb{Q}(\sqrt{-1}, \sqrt{-d})$ for solving the question of sum of two integral squares (and the idelic class group is not unique). In order to get the explicit condition for the sum of two integral squares, one needs further to construct the explicit abelian extension of $\mathbb{Q}(\sqrt{-1}, \sqrt{-d})$ corresponding to the idelic class group by class field theory. Such explicit construction is a wide open problem (Hilbert's 12th problem) in general but ad hoc method. Besides

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the ring class field of $\mathbb{Q}(\sqrt{d},\sqrt{-1})$ corresponding to the order $\mathbb{Z}+\mathbb{Z}\sqrt{-d}+\mathbb{Z}\sqrt{-1}+\mathbb{Z}\sqrt{d}$, one needs to construct some extra abelian extension of $\mathbb{Q}(\sqrt{d},\sqrt{-1})$ to test the sum of two integral squares. The author constructed such extra abelian extension for some special d when d has a prime factor which is congruent $3,5,7 \mod 8$ (see [16]). Some related works were also performed in [11] and [15]. In this paper, we consider the question over the imaginary quadratic field $\mathbb{Q}(\sqrt{-2p})$ where $p\equiv 1 \mod 8$. In the previous case, the extra abelian extension can be realized to the abelian extension of $\mathbb{Q}(\sqrt{-1})$. In this case, abelian extensions of $\mathbb{Q}(\sqrt{-1})$ will be not valid in general.

Notation and terminologies are standard if not explained. Let F be a number field, $\mathfrak{o}_F$ the ring of integers of F, Ω_F the set of all places of F and ∞ the set of all infinite places of F. For simplicity, we write $\mathfrak{p}<\infty$ for $\mathfrak{p}\in\Omega_F\setminus\infty$. Let $F_\mathfrak{p}$ be the completion of F at $\mathfrak{p}$ and $\mathfrak{o}_{F_\mathfrak{p}}$ the local completion of $\mathfrak{o}_F$ at $\mathfrak{p}$ for each $\mathfrak{p}\in\Omega_F$. Write $\mathfrak{o}_{F_\mathfrak{p}}=F_\mathfrak{p}$ for $\mathfrak{p}\in\infty$. We also denote the adele ring (resp. the idele group) of F by $\mathbb{A}_F$ (resp. $\mathbb{I}_F$).

Let $E=F(\sqrt{-1})$ and let T be the torus

$$R^1_{E/F}(\mathbb{G}_{m,E})=\mathrm{Ker}[R_{E/F}(\mathbb{G}_{m,E})\xrightarrow{N_{E/F}}\mathbb{G}_{m,F}],$$

where R denotes the Weil's restriction (see [6, p. 211]). Denote λ to be the embedding from T to $R_{E/F}(\mathbb{G}_{m,E})$. Obviously, λ induces a natural group homomorphism $\lambda_E: T(\mathbb{A}_F)\to\mathbb{I}_E$.

Let $\mathbf{X}_\alpha$ denote the scheme over $\mathfrak{o}_F$ defined by $x^2+y^2=\alpha$ for a non-zero integer $\alpha\in\mathfrak{o}_F$. Let $X_\alpha=\mathbf{X}_\alpha\times_{\mathfrak{o}_F}F$. Obviously $f=x+y\sqrt{-1}$ is an invertible function on $X_\alpha\otimes_F E$. And f induces a natural map

$$f_E: X_\alpha(\mathbb{A}_F)\to\mathbb{I}_E.$$

The restriction to $X_\alpha(F_\mathfrak{p})$ of f_E can be written by

$$f_E[(x_\mathfrak{p},y_\mathfrak{p})]=\begin{cases}(x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},x_\mathfrak{p}-y_\mathfrak{p}\sqrt{-1}), & \text{if } \mathfrak{p} \text{ splits in } E/F,\\ x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1}, & \text{otherwise.}\end{cases}$$

Definition 0.1. Let $K_1,\dots,K_n$ be finite abelian extensions over E. Let

$$\psi_{K_i/E}:\mathbb{I}_E\to\mathrm{Gal}(K_i/E)\quad \text{for } 1\leqslant i\leqslant n$$

be the Artin map. We say that α satisfies the *Artin condition of* $K_1,\dots,K_n$ *if there is*

$$\prod_{\mathfrak{p}\leqslant\infty}(x_\mathfrak{p},y_\mathfrak{p})\in\prod_{\mathfrak{p}\leqslant\infty}\mathbf{X}_\alpha(\mathfrak{o}_{F_\mathfrak{p}})$$

such that

$$\psi_{K_i/E}\Big(f_E\Big[\prod_{\mathfrak{p}\leqslant\infty}(x_\mathfrak{p},y_\mathfrak{p})\Big]\Big)=1_i\quad \textit{for } i=1,\dots,n,$$

where 1_i *is the identity element of* $\mathrm{Gal}(K_i/E)$.

By class field theory, it is a necessary condition for $\mathbf{X}_\alpha(\mathfrak{o}_F)\neq\emptyset$ that α satisfies the Artin condition of $K_1,\dots,K_n$. And there is a (not unique) finite abelian extension K/E independent of α, such that the Artin condition of K is also sufficient for $\mathbf{X}_\alpha(\mathfrak{o}_F)\neq\emptyset$ (see [17, Theorem 1.10]). Let $L=\mathfrak{o}_F+\mathfrak{o}_F\sqrt{-1}$ and H_L be the ring class field corresponding to the order L. For example, the Artin condition of H_L is sufficient for $\mathbf{X}_\alpha(\mathfrak{o}_F)\neq\emptyset$ if $F=\mathbb{Q}(\sqrt{\pm p})$ and p is a prime (see [14]). However, the Artin condition of H_L is not always sufficient for general quadratic fields.

Let $\mathbf{T}$ be the group scheme over $\mathfrak{o}_F$ defined by $x^2+y^2=1$, which is an integral model of T. Since $\mathbf{T}$ is separated over $\mathfrak{o}_F$, we can view $\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})$ as a subgroup of $T(F_\mathfrak{p})$. Furthermore, the following result can be found in [17, Corollary 1.6].

Proposition 0.2. *Let* $K_1,\dots,K_n$ *be finite abelian extensions of* E *such that the group homomorphism induced by* λ_E,

$$\widetilde{\lambda}_E:{}^{T(\mathbb{A}_F)}\!/_{T(F)\prod_{\mathfrak{p}\leqslant\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})}\longrightarrow\prod_{i=1}^n{}^{\mathbb{I}_E}\!/_{E^*N_{K_i/E}(\mathbb{I}_{K_i})}$$

is well-defined and injective, where well-defined means

$$\lambda_E\Big(T(F)\prod_{\mathfrak{p}\leqslant\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})\Big)\subset\bigcap_{i=1}^{n}(E^*N_{K_i/E}(\mathbb{I}_{K_i})).$$

Then $\mathbf{X}_\alpha(\mathfrak{o}_F)\neq\emptyset$ *if and only if* α *satisfies the Artin condition of* $K_1,\ldots,K_n$.

Let p be a prime and $F=\mathbb{Q}(\sqrt{-2p})$. If $p\equiv 5 \mod 8$, then $x^2+y^2=-1$ is solvable over $\mathfrak{o}_F$. Therefore the equation $x^2+y^2=\alpha$ is solvable over $\mathfrak{o}_F$ if and only if α satisfies the Artin condition of H_L by [14, Proposition 2.1]. If $p\equiv 3 \mod 4$, then the equation $x^2+y^2=\alpha$ is solvable over $\mathfrak{o}_F$ if and only if α satisfies the Artin condition of H_L and $E(\sqrt[4]{2})$ by [16, Proposition 2.10]. In this paper, we mainly prove the following result:

Theorem 0.3. *Let* $p\equiv 1 \mod 8$ *be a prime satisfying* $2p=r^2+s^2$ *for two integers* $r,s\equiv\pm 3 \mod 8$. *Let* $F=\mathbb{Q}(\sqrt{-2p})$ *and* $E=F(\sqrt{-1})$. *Then the diophantine equation* $x^2+y^2=\alpha$ *is solvable over* $\mathfrak{o}_F$ *if and only if* α *satisfies the Artin condition of* H_E *and* Θ, *where* H_E *is the Hilbert class field of* E *and* Θ *will be defined in Section* 1.

As an application of Theorem 0.3, we have the following result.

Proposition 0.4. *Let* p *be a prime as in Theorem* 0.3 *and* $F=\mathbb{Q}(\sqrt{-2p})$. *Then*

(1) *The equation* $x^2+y^2=-1$ *is not solvable over* $\mathfrak{o}_F$;

(2) *The equation* $x^2+y^2=-2$ *is solvable over* $\mathfrak{o}_F$ *if and only if* $(\frac{2}{p})_4=1$.

1 The construction of the field Θ

Let p be a prime with $p\equiv 1 \mod 8$. Obviously $x^2+y^2=p$ is solvable over $\mathbb{Z}$. We can choose one solution (r_1,s_1) with r_1 being odd and $4\mid s_1$. Let $F=\mathbb{Q}(\sqrt{-2p})$ and $E=F(\sqrt{-1})$. And let $H'=E(\sqrt{r_1+s_1\sqrt{-1}})$.

Proposition 1.1. H'/E *is a quadratic extension and unramified everywhere.*

Proof. First we claim that $r_1+s_1\sqrt{-1}$ is not a square in E. Otherwise $r_1+s_1\sqrt{-1}\in E^{*2}$, which implies

$$p=N_{E/F}(r_1+s_1\sqrt{-1})\in F^{*2}.$$

It is impossible since p is not a square in F. Therefore H'/E is a quadratic extension.

Obviously, H'/E is unramified at any place $v\nmid 2p$. Let v_1,v_2 be the two places of E above p. Since r_1 and s_1 are relatively prime, we have $\mathrm{ord}_{v_i}(r_1+s_1\sqrt{-1})$ is 0 or 2 for $i=1,2$. Therefore H'/E is unramified at v_1 and v_2. Let v be the unique place of E above 2. Since $r_1\equiv 1 \mod 2$ and $s_1\equiv 0 \mod 4$, we have

$$r_1+s_1\sqrt{-1}\equiv 1 \text{ or } 5 \quad \text{in } E_v^*/E_v^{*2}.$$

Therefore H'/E is unramified at v. □

In order to construct Θ, we need to consider the solvability of the Diophantine equation

$$x^2-(r_1+s_1\sqrt{-1})y^2=(r_1\sqrt{-1}+s_1)z^2. \tag{1.1}$$

The field Θ will be constructed by a non-trivial solution of (1.1).

Lemma 1.2. *The equation* (1.1) *has a non-trivial solution in* $\mathbb{Z}[\sqrt{-1}]$.

Proof. By Hasse principle we only need to show the equation (1.1) is locally solvable. Therefore we only need to show

$$(r_1+s_1\sqrt{-1},r_1\sqrt{-1}+s_1)_v=1$$

for all $v\mid 2p$. Since $(r_1+s_1\sqrt{-1})(r_1\sqrt{-1}+s_1)=p\sqrt{-1}$ and $2\sqrt{-1}=(1+\sqrt{-1})^2$, we have

$$(r_1+s_1\sqrt{-1},r_1\sqrt{-1}+s_1)_v=(r_1+s_1\sqrt{-1},2p)_v.$$

(i) Let v be the unique place of $\mathbb{Q}(\sqrt{-1})$ above 2. One obtains

$$(r_1 + s_1\sqrt{-1}, 2p)_v = (p, 2p)_2 = 1$$

since $p \equiv 1 \mod 8$.

(ii) Suppose $v \mid p$. Then $\mathbb{Q}(\sqrt{-1})$ is split at v. And $\sqrt{-1} \equiv \pm r_1/s_1 \mod p$. So we have $r_1 + s_1\sqrt{-1} \equiv 0$ or $2r_1 \mod p$. Since r_1 and s_1 are relatively prime, we have

$$\mathrm{ord}_v(r_1 + s_1\sqrt{-1}) = 0 \text{ or } 1.$$

Since $(r_1 + s_1\sqrt{-1})(r_1 - s_1\sqrt{-1}) = p$ and $(p, 2p)_p = 1$, we only need to show

$$(r_1 + s_1\sqrt{-1}, 2p)_v = 1 \quad \text{if } \mathrm{ord}_v(r_1 + s_1\sqrt{-1}) = 0.$$

Therefore we can assume $r_1 + s_1\sqrt{-1} \equiv 2r_1 \mod p$. Then

$$(r_1 + s_1\sqrt{-1}, 2p)_v = (2r_1, p)_p = (r_1, p)_p.$$

By Hilbert's reciprocity law, we have

$$(r_1, p)_p = (r_1, p)_2 \prod_{l \mid r_1} (r_1, p)_l = 1 \cdot 1 = 1$$

since $p \equiv 1 \mod 8$ and $r_1^2 + s_1^2 = p$. Therfore one has $(r_1 + s_1\sqrt{-1}, 2p)_v = 1$. □

Since the ring $\mathbb{Z}(\sqrt{-1})$ is a PID, we can choose a nonzero solution (x_0, y_0, z_0) of the equation (1.1) such that x_0, y_0 and z_0 have no common divisor. Define

$$\Theta = E(\sqrt{x_0 + y_0\sqrt{r_1 + s_1\sqrt{-1}}}).$$

Obviously $\Theta \supset H'$.

Proposition 1.3. *Θ/E is cyclic of order 4 and unramified everywhere except at the unique palce above 2.*

Proof. Note that the valuation of $r_1\sqrt{-1} + s_1$ is even at any finite place of E except above 2. By the choice of (x_0, y_0, z_0), we know x_0 and y_0 are relatively prime. So the valuation of $x_0 + y_0\sqrt{r_1 + s_1\sqrt{-1}}$ is also even at at any finite place of E except above 2. Then Θ/E is unramified everywhere except at the unique palce above 2.

Since $(r_1 + s_1\sqrt{-1})(r_1\sqrt{-1} + s_1) = (\sqrt{2p}(1 + \sqrt{-1}))^2/2^2$, we have $r_1\sqrt{-1} + s_1$ is a square in H'. Then we have Θ/E is Galois. Assume $\mathrm{Gal}(\Theta/E) \neq \mathbb{Z}/4\mathbb{Z}$, then there exist $a \in E^*$ and $b \in H'^*$ such that $x_0 + y_0\sqrt{r_1 + s_1\sqrt{-1}} = ab^2$ (if $\mathrm{Gal}(\Theta/E) = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ then a is not a square in H'^*). So we have $(r_1\sqrt{-1} + s_1)z_0^2 = a^2 N_{H'/E}(b)^2$. So we have $r_1\sqrt{-1} + s_1 \in E^{*2}$, which contradicts Proposition 1.1. □

2 Proof of Theorem 0.3

Let $F = \mathbb{Q}(\sqrt{-2p})$ and $\mathfrak{o}_F$ be the ring of integers of F. Let $E = F(\sqrt{-1})$ and let Θ be defined as in Section 1. Write $\mathrm{Gal}(\Theta/E) = \{\pm 1, \pm \mathrm{i}\}$ here $\mathrm{i}^2 = 1$. In order to prove Theorem 0.3, we need the following two propositions.

Proposition 2.1. *Let $p \equiv 1 \mod 8$ be a prime. If x_2 and y_2 in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfy $x_2^2 + y_2^2 = 1$, then the Artin symbol*

$$\psi_{\Theta/E,v}(x_2 + y_2\sqrt{-1}) = 1$$

where v (resp. $\mathfrak{p}$) is the unique place of E (resp. F) above 2.

Before proving this proposition, we give the following two lemmas:

Lemma 2.2. *Let v (resp. $\mathfrak{p}$) be the unique place of E (resp. F) above 2. Let x_2 and y_2 in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfy $x_2^2+y_2^2=1$. Then there exists $\beta\in\mathfrak{o}_{E_v}^\times$ such that $x_2+y_2\sqrt{-1}=\sigma(\beta)/\beta$, where σ is the non-trivial element of* $\mathrm{Gal}(E_v/F_\mathfrak{p})$.

Proof. Let $L=\mathfrak{o}_F+\mathfrak{o}_F\sqrt{-1}\subset\mathfrak{o}_E$. The order L is not the maximal order of E since $(\sqrt{2p}+\sqrt{-2p})/2\notin L$. And the maximal order $\mathfrak{o}_E=L[(\sqrt{2p}+\sqrt{-2p})/2]$. It is easy to see that $\sqrt{-2p}\cdot\alpha$ and $\alpha^2\in L$ for any $\alpha\in\mathfrak{o}_E$.

By Hilbert's theorem 90, there exists $\beta\in E_v^*$ such that $x_2+y_2\sqrt{-1}=\sigma(\beta)/\beta$. Without loss of generality, we can assume $\beta\in\mathfrak{o}_{E_v}$ with $\mathrm{ord}_v(\beta)=0$ or 1.

If $\beta\in\mathfrak{o}_{E_v}^\times$, then $\sigma(\beta)/\beta=\sigma(\beta)^2/N_{E_v/F_\mathfrak{p}}(\beta)$. Let $L_\mathfrak{p}=L\otimes_{\mathfrak{o}_F}\mathfrak{o}_{F_\mathfrak{p}}$ and $L_\mathfrak{p}^\times$ the set consisting of all invertible elements of $L_\mathfrak{p}$. Since $\mathfrak{o}_F\subset L$ and $\alpha^2\in L$ for any $\alpha\in\mathfrak{o}_E$, we have $\sigma(\beta)/\beta\in L_\mathfrak{p}^\times$.

In the following, we will show $\mathrm{ord}_v(\beta)\neq 1$. Otherwise we can write $\beta=(\xi_8-1)\beta'$ where $\xi_8=(\sqrt{-2}+\sqrt{2})/2$ is a primitive 8-th root of unity and β' is a unit in $\mathfrak{o}_{E_v}$, since $E_v=\mathbb{Q}_2(\xi_8)$ and ξ_8-1 is a uniformizer of E_v. By the above argument we know $\sigma(\beta')/\beta'\in L_\mathfrak{p}^\times$. Hence we have $\sigma(\xi_8-1)/(\xi_8-1)\in L_\mathfrak{p}$. Since $\sigma(\xi_8)=(\sqrt{-2}-\sqrt{2})/2=-\xi_8^{-1}$, one has

$$-1/\xi_8-2\xi_8^{-1}/(\xi_8-1)=\sigma(\xi_8-1)/(\xi_8-1)\in L_\mathfrak{p}.$$

Since $\sqrt{2}\xi_8^{-1}/(\xi_8-1)\in\mathfrak{o}_{E_v}$, one obtains $2\xi_8^{-1}/(\xi_8-1)=\sqrt{2}\cdot\sqrt{2}\xi_8^{-1}/(\xi_8-1)\in L_\mathfrak{p}$. Therefore $-1/\xi_8\in L_\mathfrak{p}$, which is impossible. □

Lemma 2.3. *Let p be a prime with $p\equiv 1 \mod 8$. Let $F=\mathbb{Q}(\sqrt{-2p})$. Let $K=\mathbb{Q}(\sqrt{-1})$ and $E=F(\sqrt{-1})$. If $\alpha\in\mathfrak{o}_{K_w}^\times$, then the Artin symbol*

$$\psi_{\Theta/E,v}(\alpha)=1$$

where v (resp. w) is the unique place of E (resp. K) above 2.

Proof. Let $K'=K(\sqrt{r_1+s_1\sqrt{-1}})$, where r_1 and s_1 are defined as in Section 1. We can see K'/K is unramified at w by a similar argument as in Proposition 1.1.

If K'/K is split at w, then

$$\psi_{\Theta/E,v}(\alpha)=(\alpha,x_0+y_0\sqrt{r_1+s_1\sqrt{-1}})_v=1$$

since $[E_v:K_w]=2$ and $\alpha,x_0+y_0\sqrt{r_1+s_1\sqrt{-1}}\in K_w$, where x_0 and y_0 are defined as in Section 1.

If K'/K is inert at w, then there exists $\alpha'\in K'^*$ such that $N_{K'_{w'}/K_w}(\alpha')=\alpha$, where w' is the unique place of K' above w. Recall $H'=E(\sqrt{r_1+s_1\sqrt{-1}})$ and $K'\subset H'$. Let $\mathfrak{w}$ be the unique place of H' above w'. Then

$$N_{K'_{w'}/K_w}(\alpha')=N_{H'_\mathfrak{w}/E_v}(\alpha').$$

Since $K'_{w'}/K_w$ is unramified and E_v/K_w is ramified, one obtains $K'_{w'}\cap E_v=K_w$. Therefore $[H'_\mathfrak{w}:K'_{w'}]=2$. By the local class field theory, one has

$$\begin{aligned}\psi_{\Theta/E,v}(\alpha)&=\psi_{\Theta/E,v}(N_{K'_{w'}/K_w}(\alpha'))=\psi_{\Theta/E,v}(N_{H'_\mathfrak{w}/E_v}(\alpha'))\\&=(\alpha',x_0+y_0\sqrt{r_1+s_1\sqrt{-1}})_\mathfrak{w}=1\end{aligned}$$

since $[H'_\mathfrak{w}:K'_{w'}]=2$ and $\alpha',x_0+y_0\sqrt{r_1+s_1\sqrt{-1}}\in K'_{w'}$. □

Proof of Proposition 2.1. By Lemma 2.2, there exists $\beta\in\mathfrak{o}_{E_v}^\times$ such that $x_2+y_2\sqrt{-1}=\sigma(\beta)/\beta$. Then we have

$$\psi_{\Theta/E,v}(x_2+y_2\sqrt{-1})=\psi_{\Theta/E,v}(N_{E_v/F_\mathfrak{p}}(\beta))\cdot\psi_{\Theta/E,v}(\beta^2)^{-1}.$$

The proof follows $\psi_{\Theta/E,v}(\beta^2)=1$ and $\psi_{\Theta/E,v}(N_{E_v/F_\mathfrak{p}}(\beta))=1$.

(1) The Artin symbol

$$\psi_{\Theta/E,v}(\beta^2)=(\beta,r_1+s_1\sqrt{-1})_v$$

since Θ/E is cyclic of order 4. Let $H' = E(\sqrt{r_1 + s_1\sqrt{-1}})$. By Lemma 1.1, H'/E is unramified everywhere. Therefore $(\beta, r_1 + s_1\sqrt{-1})_v = 1$ since β is a unit.

(2) Since $E_v = \mathbb{Q}_2(\sqrt{-1}, \sqrt{-2})$ and β is a unit, we have

$$N_{E_v/\mathbb{Q}_2}(\beta) \equiv 1 \bmod 8.$$

Then there exists $s \in \mathbb{Z}_2^\times$ such that $N_{E_v/\mathbb{Q}_2}(\beta) = s^2$. By Hilbert's theorem 90, there exists $u \in F_{\mathfrak{p}}^*$ such that

$$N_{E_v/F_{\mathfrak{p}}}(\beta) = s\sigma'(u)/u,$$

where σ' is the non-trivial element of $\mathrm{Gal}(F_{\mathfrak{p}}/\mathbb{Q}_2)$. So

$$\psi_{\Theta/E,v}(N_{E_v/F_{\mathfrak{p}}}(\beta)) = \psi_{\Theta/E,v}(sN_{F_{\mathfrak{p}}/\mathbb{Q}_2}(u)) \cdot \psi_{\Theta/E,v}(u^2)^{-1}.$$

We have $\psi_{\Theta/E,v}(u^2) = (u, r_1 + s_1\sqrt{-1})_v = (u,p)_{\mathfrak{p}} = 1$ since $p \equiv 1 \mod 8$. Therefore we only need to show

$$\psi_{\Theta/E,v}(sN_{F_{\mathfrak{p}}/\mathbb{Q}_2}(u)) = 1.$$

Let $K = \mathbb{Q}(\sqrt{-1})$ and w the unique place of K above 2. Since $sN_{F_{\mathfrak{p}}/\mathbb{Q}_2}(u) \in \mathbb{Q}_2^\times$, one has

$$sN_{F_{\mathfrak{p}}/\mathbb{Q}_2}(u) = (1+\sqrt{-1})^{2i}\alpha,$$

where $i \in \mathbb{Z}$ and $\alpha \in \mathfrak{o}_{K_w}^\times$. By Lemma 2.3, one has $\psi_{\Theta/E,v}(\alpha) = 1$. Since $\mathrm{ord}_v(1+\sqrt{-1}) = 2$, one obtains

$$\psi_{\Theta/E,v}((1+\sqrt{-1})^{2i}) = ((1+\sqrt{-1})^i, r_1 + s_1\sqrt{-1})_v = 1$$

since $[E_v : K_w] = 2$ and $1+\sqrt{-1}, r_1 + s_1\sqrt{-1} \in K_w$. Therefore $\psi_{\Theta/E,v}(sN_{F_{\mathfrak{p}}/\mathbb{Q}_2}(u)) = 1$. □

Proposition 2.4. *Let p be a prime as in Theorem* 0.3. *If x_2 and y_2 in $\mathfrak{o}_{F_{\mathfrak{p}}}$ satisfy $x_2^2 + y_2^2 = -1$, then the Artin symbol*

$$\psi_{\Theta/E,v}(x_2 + y_2\sqrt{-1}) = -1,$$

where v (resp. $\mathfrak{p}$) is the unique place in E (resp. F) above 2.

Proof. Let $\eta = (s - \sqrt{2p})/r$. By Proposition 2.1, one has

$$\psi_{\Theta/E,v}(x_2 + y_2\sqrt{-1}) = \psi_{\Theta/E,v}(\eta)$$

since $(x_2 + y_2\sqrt{-1})/\eta \in \mathfrak{o}_{F_{\mathfrak{p}}} + \mathfrak{o}_{F_{\mathfrak{p}}}\sqrt{-1}$ and its norm is 1. By Lemma 2.3, one has $\psi_{\Theta/E,v}(r) = 1$. The proof follows if we show

$$\psi_{\Theta/E,v}(s - \sqrt{2p}) = -1.$$

By global class field theory, we have

$$\psi_{\Theta/E,v}(s - \sqrt{2p}) = \prod_{w|r} \psi_{\Theta/E,w}(s - \sqrt{2p}).$$

For any prime $l \mid r$, l splits into two places $v_{l,1}$ and $v_{l,2}$ of $\mathbb{Q}(\sqrt{2p})$. Since r and s are relatively prime, we have

$$\mathrm{ord}_{v_{l,1}}(s - \sqrt{2p}) \cdot \mathrm{ord}_{v_{l,2}}(s - \sqrt{2p}) = 0.$$

Without loss of generality, we can assume

$$\mathrm{ord}_{v_{l,1}}(s - \sqrt{2p}) = \mathrm{ord}_{v_{l,1}}(r^2) \quad \text{and} \quad \mathrm{ord}_{v_{l,2}}(s - \sqrt{2p}) = 0.$$

Since Θ/E is unramified everywhere except at v by Proposition 1.3, we have

$$\prod_{w|l} \psi_{\Theta/E,w}(s - \sqrt{2p}) = \prod_{w|v_{l,1}} \psi_{\Theta/E,w}(r^2) = \prod_{w|v_{l,1}} (r, r_1 + s_1\sqrt{-1})_w = (r,p)_l.$$

Therefore,

$$\psi_{\Theta/E,v}(s-\sqrt{2p})=\prod_{l|r}(p,r)_l=(p,r)_2\cdot(p,r)_p=(p,r)_p.$$

Since $r^2+s^2=2p$, one has $(2p,r)_l=1$ for any prime $l\mid r$. Then

$$(p,r)_p=(2p,r)_p=(2p,r)_2\cdot\prod_{l|r}(2p,r)_l=(2,r)_2\cdot 1=-1$$

since $r\equiv\pm 3 \mod 8$. Then we have

$$\psi_{\Theta/E,v}(s-\sqrt{2p})=-1.$$ □

Corollary 2.5. *Let p be a prime as in Theorem* 0.3. *Let $F=\mathbb{Q}(\sqrt{-2p})$. Then the equation $x^2+y^2=-1$ is not solvable over $\mathfrak{o}_F$.*

Proof. Let $\mathbf{X}_{(\alpha)}$ denote the affine scheme defined by $x^2+y^2=\alpha$ for $\alpha\in\mathfrak{o}_F$. Let $E=\mathbb{Q}(\sqrt{-1},\sqrt{-2p})$ and let Θ be given as in Section 1. Let v be the unique place of E above 2. By Proposition 2.4, one has

$$\psi_{\Theta/E,v}(x_2+y_2\sqrt{-1})=-1$$

for any $x_2,y_2\in\mathfrak{o}_{F_\mathfrak{p}}$ with $x_2^2+y_2^2=-1$, where $\mathfrak{p}$ is the unique place of F above 2. By Proposition 1.3, Θ/E is unramified everywhere except at v. This implies that

$$\psi_{\Theta/E}\Big(f_E\Big[\prod_{\mathfrak{p}\leqslant\infty}(x_\mathfrak{p},y_\mathfrak{p})\Big]\Big)=\psi_{\Theta/E,v}(x_2+y_2\sqrt{-1})=-1$$

for any $\prod_{\mathfrak{p}\leqslant\infty}(x_\mathfrak{p},y_\mathfrak{p})\in\prod_{\mathfrak{p}\leqslant\infty}\mathbf{X}_{(-1)}(\mathfrak{o}_{F_\mathfrak{p}})$. Then $x^2+y^2=-1$ is not solvable over $\mathfrak{o}_F$ by class field theory. □

Lemma 2.6. *Let d be a square-free positive odd number. Let $F=\mathbb{Q}(\sqrt{2d})$ or $\mathbb{Q}(\sqrt{-2d})$ and let $\mathfrak{o}_F$ be the ring of integers of F. Let $L=\mathfrak{o}_F+\mathfrak{o}_F\sqrt{-1}$ and H_L the ring class field corresponding to L. Let $E=F(\sqrt{-1})$ and H_E the Hilbert class field of E. If one of the equations $x^2-2dy^2=\pm 2$ is solvable over $\mathbb{Z}$, then*

$$H_L=H_E.$$

Proof. Let v and $\mathfrak{p}$ be respectively the unique place of E and F above 2. Let $L_\mathfrak{p}=L\otimes_{\mathfrak{o}_F}\mathfrak{o}_{F_\mathfrak{p}}$. Let $\mathfrak{o}_E$ be the ring of integers of E. And the maximal order

$$\mathfrak{o}_E=L[(\sqrt{2d}+\sqrt{-2d})/2].$$

By [9, VI. §1, Exercise 13], we have the following exact sequence

$$1\to\mathfrak{o}_E^\times/L^\times\to\mathfrak{o}_{E_v}^\times/L_\mathfrak{p}^\times\to Cl_L\to Cl_E\to 1,$$

where $Cl_E\cong\mathrm{Gal}(H_E/E)$ and $Cl_L\cong\mathrm{Gal}(H_L/E)$ are class groups. It is easy to verify that $\mathfrak{o}_{E_v}^\times/L_\mathfrak{p}^\times$ is of order 2 since $(1+\sqrt{-1})\alpha,\alpha^2\in L$ for any $\alpha\in\mathfrak{o}_E$. Since one of the equations $x^2-2dy^2=\pm 2$ is solvable over $\mathbb{Z}$, we can choose one solution (x_0,y_0). Let

$$\eta=(x_0-y_0\sqrt{2d})/(1+\sqrt{-1}).$$

It is easy to see

$$\mathrm{ord}_v(\eta)=\mathrm{ord}_v(x_0-y_0\sqrt{2d})-\mathrm{ord}_v(1+\sqrt{-1})=2-2=0.$$

And $\mathrm{ord}_w(\eta)=0$ for any place of E not above 2. Therefore η is a unit of $\mathfrak{o}_E$. Since x_0 is even and y_0 is odd, one obtains $\eta\notin L$. Then we have that $|\mathfrak{o}_E^\times/L^\times|\geqslant 2$. So the morphism $\mathfrak{o}_E^\times/L^\times\to\mathfrak{o}_{E_v}^\times/L_\mathfrak{p}^\times$ is surjective. So we have $Cl_L=Cl_E$. □

Let p be a prime, we know that one of the three equations

$$x^2 - 2py^2 = -1, \quad x^2 - 2py^2 = \pm 2$$

is solvable over $\mathbb{Z}$ (see [2, p. 225]). Let p be a prime as in Theorem 0.3. Then $x^2 - 2py^2 = -1$ is not solvable over $\mathbb{Z}$ (see [3, 12] or [17, Corollary 5.3]). Therefore one of the equations $x^2 - 2py^2 = \pm 2$ is solvable over $\mathbb{Z}$. By using Lemma 2.6 we have the following corollary.

Corollary 2.7. *Let p be a prime as in Theorem* 0.3. *Let $F = \mathbb{Q}(\sqrt{2p})$ or $\mathbb{Q}(\sqrt{-2p})$ and $\mathfrak{o}_F$ the ring of integers of F. Let $L = \mathfrak{o}_F + \mathfrak{o}_F\sqrt{-1}$ and H_L the ring class field corresponding to L. Let $E = F(\sqrt{-1})$ and H_E the Hilbert class field of E. Then*

$$H_L = H_E.$$

Proof of Theorem 0.3. Let $\mathfrak{p}$ be a place of F and $L_\mathfrak{p} = L \otimes_{\mathfrak{o}_F} \mathfrak{o}_{F_\mathfrak{p}}$. Recall $T = R^1_{E/F}(\mathbb{G}_{m,E})$ and $\mathbf{T}$ is the scheme defined by the equation $x^2 + y^2 = 1$, we have

$$T(F) = \{\beta \in E^* : \ N_{E/F}(\beta) = 1\}$$

and

$$\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}}) = \{\beta \in L_\mathfrak{p}^\times : \ N_{E_\mathfrak{p}/F_\mathfrak{p}}(\beta) = 1\}.$$

And $L_\infty^\times = E_\infty^* = \mathbb{C}^* \times \mathbb{C}^*$.

Let v and $\mathfrak{p}$ be respectively the unique places of E and F above 2. Let λ_E be the natural group homomorphism

$$\lambda_E : T(\mathbb{A}_F) \to \mathbb{I}_E.$$

By Proposition 2.1, one obtains

$$\lambda_E(\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})) \subseteq N_{\Theta_v/E_v}(\Theta_v^*),$$

where $\Theta_v = \Theta \otimes_E E_v$. Since Θ/E is unramified everywhere except at v, one obtains the natural group homomorphism

$$\widetilde{\lambda}_E : T\Big(\mathbb{A}_F)/(T(F)\prod_{\mathfrak{p}\leqslant\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})\Big) \longrightarrow [\mathbb{I}_E/(E^*N_{\Theta/E}(\mathbb{I}_\Theta))] \times \Big[\mathbb{I}_E\Big/\Big(E^*\prod_{v\leqslant\infty}\mathfrak{o}_{E_v}^\times\Big)\Big]$$

is well-defined. By Proposition 0.2, we only need to show $\widetilde{\lambda}_E$ is injective.

Let $u \in \operatorname{Ker}\widetilde{\lambda}_E$. By Corollary 2.7, we have

$$E^*\prod_{\mathfrak{p}\leqslant\infty} L_\mathfrak{p}^\times = E^*\prod_{v\leqslant\infty}\mathfrak{o}_{E_v}^\times.$$

Then there are $\beta \in E^*$ and $i \in \prod_{\mathfrak{p}\leqslant\infty} L_\mathfrak{p}^\times$ with $\lambda_E(u) = \beta i$. We have

$$N_{E/F}(\beta) = N_{E/F}(i)^{-1} \in F^* \cap \prod_{\mathfrak{p}\leqslant\infty}\mathfrak{o}_{F_\mathfrak{p}}^\times = \{\pm 1\},$$

since F is an imaginary quadratic field and $F \neq \mathbb{Q}(\sqrt{-1})$ or $\mathbb{Q}(\sqrt{-3})$.

If $N_{E/F}(\beta) \neq 1$, one obtains $N_{E/F}(\beta) = N_{E/F}(i) = -1$. Write $i = (i_w)_w \in \mathbb{I}_E, w \in \Omega_E$. Since Θ/E is unramified everywhere except at v, one concludes that $\psi_{\Theta/E}(i_w)$ is trivial for all places $w \neq v$, where i_w is regarded as an idele whose w-component is i_w and 1 otherwise. Since $N_{E/F}(i_v) = -1$, one gets

$$\psi_{\Theta/E}(\alpha i) = \psi_{\Theta/E}(i) = \psi_{\Theta/E}(i_v) = -1$$

by Proposition 2.4, where $\psi_{\Theta/E}$ is the Artin map. This contradicts $u \in \ker\widetilde{\lambda}_E$.

Therefore $N_{E/F}(\beta) = 1$, one concludes that

$$N_{E/F}(\beta) = N_{E/F}(i) = 1 \quad \Rightarrow \quad \beta \in T(F) \quad \text{and} \quad i \in \prod_{\mathfrak{p}\leqslant\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}}).$$

So $\beta i \in T(F)\prod_{\mathfrak{p}\leqslant\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})$. □

3 Some applications

Let $F=\mathbb{Q}(\sqrt{-2p})$, $\mathfrak{o}_F$ be the ring of integers of F and $E=F(\sqrt{-1})$. Let Θ be defined as in Section 1.

Lemma 3.1. *Let p be a prime as in Theorem* 0.3. *Let v (resp. $\mathfrak{p}$) be the unique place of E (resp. F) above* 2. *Let $\mathbf{X}_{-2}$ be the affine scheme over $\mathfrak{o}_F$ defined by $x^2+y^2=-2$. Then*

$$\left\{\left(\frac{2}{p}\right)_4\right\}\subset\psi_{\Theta/E,v}(f_E[\mathbf{X}_{-2}(\mathfrak{o}_{F_\mathfrak{p}})])\subset\left\{-1,\left(\frac{2}{p}\right)_4\right\}.$$

Proof. Let the order $L=\mathfrak{o}_F+\mathfrak{o}_F\sqrt{-1}\subset\mathfrak{o}_E$ and $L_\mathfrak{p}=\mathfrak{o}_{F_\mathfrak{p}}+\mathfrak{o}_{F_\mathfrak{p}}\sqrt{-1}\subset\mathfrak{o}_{E_v}$. Let

$$\eta_0=(s-\sqrt{2p})\cdot(1+\sqrt{-1})/r.$$

Obviously, $\eta_0\in L_\mathfrak{p}$ and $N_{E_v/F_\mathfrak{p}}(\eta_0)=-2$. Let x_2 and y_2 in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfy $x_2^2+y_2^2=-2$. And $N_{E_v/F_\mathfrak{p}}[(x_2+y_2\sqrt{-1})/\eta_0]=1$. By Hilbert's theorem 90, there is $\beta\in E_v^*$ such that

$$(x_2+y_2\sqrt{-1})/\eta_0=\sigma(\beta)/\beta,$$

where σ is the non-trivial element of $\mathrm{Gal}(E_v/F_\mathfrak{p})$. Since $E_v/F_\mathfrak{p}$ is ramified and of degree 2, we can choose $\beta\in\mathfrak{o}_{F_\mathfrak{p}}$ with $\mathrm{ord}_v(\beta)=0$ or 1.

(1) Suppose $\mathrm{ord}_v(\beta)=0$. Then $\sigma(\beta)/\beta\in L_\mathfrak{p}$. By Propositions 2.1 and 2.4, we have

$$\psi_{\Theta/E,v}(\sigma(\beta)/\beta)=1\quad\text{and}\quad\psi_{\Theta/E,v}((s-\sqrt{2p})/r)=-1.$$

Since Θ/E is unramified everywhere except at v, we have

$$\psi_{\Theta/E,v}(1+\sqrt{-1})=\prod_{w\neq v}\psi_{\Theta/E,w}(1+\sqrt{-1})=1.$$

This implies

$$\psi_{\Theta/E,v}(x_2+y_2\sqrt{-1})=\psi_{\Theta/E,v}(\eta_0)=-1.$$

(2) In the following we consider the case $\mathrm{ord}_v(\beta)=1$.

Since $p\equiv 1 \mod 8$, we have $x^2+2y^2=2p$ is solvable over $\mathbb{Z}$ by the local-global principle. Choose one solution (a,b) of this equation and let $\eta=(a-\sqrt{2p})/b$. It is easy to see that $2\mid a$ and b is odd. Then $\eta\in L_\mathfrak{p}$ and $N_{E_v/F_\mathfrak{p}}(\eta)=-2$. Since $E_v/F_\mathfrak{p}$ is ramified and of degree 2, we can choose $\beta_\eta\in\mathfrak{o}_{F_\mathfrak{p}}$ with $\mathrm{ord}_v(\beta_\eta)=0$ or 1 such that $\eta/\eta_0=\sigma(\beta_\eta)/\beta_\eta$. And

$$\eta/\eta_0=(a(1-\sqrt{-1})/2-(\sqrt{2p}-\sqrt{-2p})/2)\cdot(s+\sqrt{2p})/rb.$$

Note that $(s+\sqrt{2p})/rb\in L_\mathfrak{p}^\times$, $a(1-\sqrt{-1})/2\in L_\mathfrak{p}$ and $(\sqrt{2p}-\sqrt{-2p})/2\notin L_\mathfrak{p}$. So $\eta/\eta_0\notin L_\mathfrak{p}$. Therefore $\mathrm{ord}_v(\beta_\eta)=1$, since otherwise $\mathrm{ord}_v(\beta_\eta)=0$. Then we have $\eta/\eta_0\in L_\mathfrak{p}$, this contradicts $\eta/\eta_0\notin L_\mathfrak{p}$.

Let x_2 and y_2 in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfy $x_2^2+y_2^2=-2$, and $(x_2+y_2\sqrt{-1})/\eta_0=\sigma(\beta)/\beta$ with $\mathrm{ord}_v(\beta)=1$. Then we have $(x_2+y_2\sqrt{-1})/\eta=\sigma(\mu)/\mu$, where $\mu=\beta/\beta_\eta$. So $\mathrm{ord}_v(\mu)=0$. Then $(x_2+y_2\sqrt{-1})/\eta\in L_\mathfrak{p}^\times$ and its norm is 1. By Proposition 2.1, we have

$$\psi_{\Theta/E,v}(x_2+y_2\sqrt{-1})=\psi_{\Theta/E,v}(\eta).$$

In the following we will prove

$$\psi_{\Theta/E,v}(\eta)=\left(\frac{2}{p}\right)_4.$$

The Artin symbol

$$\psi_{\Theta/E,v}(\eta)=\psi_{\Theta/E,v}(a-\sqrt{2p})\cdot\psi_{\Theta/E,v}(b)^{-1}.$$

By Lemma 2.3, one has $\psi_{\Theta/E,v}(b)=1$. So we only need to show

$$\psi_{\Theta/E,v}(a-\sqrt{2p})=\left(\frac{2}{p}\right)_4.$$

By global class field theory, we have

$$\psi_{\Theta/E,v}(a-\sqrt{2p})=\prod_{w\neq v}\psi_{\Theta/E,w}(a-\sqrt{2p}).$$

Since $a-\sqrt{2p}$ is a unit for $w\nmid 2b$, we have $\psi_{\Theta/E,w}(a-\sqrt{2p})=1$ for $w\nmid 2b$. Then one has

$$\psi_{\Theta/E,v}(a-\sqrt{2p})=\prod_{w|b}\psi_{\Theta/E,w}(a-\sqrt{2p}).$$

For any prime $l\mid b$, l splits into two places $v_{l,1}$ and $v_{l,2}$ in $\mathbb{Q}(\sqrt{2p})$. Since $a^2-2p=-2b^2$, we have a and b are relatively prime, hence

$$\mathrm{ord}_{v_{l,1}}(a-\sqrt{2p})\cdot\mathrm{ord}_{v_{l,2}}(a-\sqrt{2p})=0.$$

Without loss of generality, we can assume

$$\mathrm{ord}_{v_{l,1}}(a-\sqrt{2p})=\mathrm{ord}_{v_{l,1}}(b^2)\quad\text{and}\quad\mathrm{ord}_{v_{l,2}}(a-\sqrt{2p})=0.$$

Since Θ/E is unramified everywhere except at v, we have

$$\prod_{w|l}\psi_{\Theta/E,w}(a-\sqrt{2p})=\prod_{w|v_{l,1}}\psi_{\Theta/E,w}(b^2)=\prod_{w|v_{l,1}}(b,r_1+s_1\sqrt{-1})_w=(b,p)_l.$$

Therefore we have

$$\psi_{\Theta/E,v}(a-\sqrt{2p})=\prod_{l|b}(p,b)_l=\left(\frac{b}{p}\right).$$

Since $a^2+2b^2=2p$, we have $\left(\frac{p}{l}\right)=1$ for any odd prime $l|a$. Therefore

$$\left(\frac{a}{p}\right)=(a,p)_2\prod_{\text{odd }l|a}(a,p)_l=1\cdot 1=1.$$

Then

$$\psi_{\Theta/E,v}(a-\sqrt{2p})=\left(\frac{b}{p}\right)=\left(\frac{b/a}{p}\right).$$

Since $a^2+2b^2=2p$, we have $(a/b)^2\equiv -2 \mod p$. Hence

$$\left(\frac{a/b}{p}\right)=\left(\frac{-2}{p}\right)_4.$$

Since $p\equiv 1 \mod 8$, we have $(\frac{-2}{p})_4=(\frac{2}{p})_4$. So we have

$$\psi_{\Theta/E,v}(a-\sqrt{2p})=\left(\frac{2}{p}\right)_4.$$

□

Proposition 3.2. *Let p be a prime as in Theorem* 0.3. *Let $F=\mathbb{Q}(\sqrt{-2p})$. Then the equation $x^2+y^2=-2$ is solvable over $\mathfrak{o}_F$ if and only if $(\frac{2}{p})_4=1$.*

Proof. Let $\mathbf{X}_{(-2)}$ denote the affine scheme defined by $x^2+y^2=-2$ for $\alpha\in\mathfrak{o}_F$. Let $E=\mathbb{Q}(\sqrt{-1},\sqrt{-2p})$ and let Θ be given as in Section 1. Let v (resp. $\mathfrak{p}$) be the unique place of E (resp. F) above 2.

First we show the sufficiency. Obviously $\mathbf{X}_{(-2)}(\mathfrak{o}_{F_\mathfrak{l}})\neq\emptyset$ for $\mathfrak{l}\leqslant\infty$. Since H_E/E is unramified everywhere, by global class field theory we have

$$\begin{aligned}1&=\prod_{w\leqslant\infty}\psi_{H_E/E,w}(1+\sqrt{-1})=\psi_{H_E/E,v}(1+\sqrt{-1})\cdot\prod_{w\neq v}\psi_{H_E/E,w}(1+\sqrt{-1})\\&=\psi_{H_E/E,v}(1+\sqrt{-1})\cdot 1=\psi_{H_E/E,v}(1+\sqrt{-1}).\end{aligned}$$

For any $(x_{\mathfrak{l}}, y_{\mathfrak{l}}) \in \mathbf{X}_{-2}(\mathfrak{o}_{F_{\mathfrak{p}}})$, obviously

$$\mathrm{ord}_v(x_{\mathfrak{l}} + y_{\mathfrak{l}}\sqrt{2p}) = \mathrm{ord}_v(1+\sqrt{-1}) = 2.$$

Since H_E is unramified at v, one obtains

$$\psi_{H_E/E,v}(x_{\mathfrak{l}} + y_{\mathfrak{l}}\sqrt{2p}) = \psi_{H_E/E,v}(1+\sqrt{-1}) = 1.$$

Therefore

$$\psi_{H_E/E}\bigg(f_E\Big[\prod_{\mathfrak{l}\leqslant\infty}(x_{\mathfrak{l}}, y_{\mathfrak{l}})\Big]\bigg) = 1$$

for any $\prod_{\mathfrak{l}\leqslant\infty}(x_{\mathfrak{l}}, y_{\mathfrak{l}}) \in \prod_{\mathfrak{l}\leqslant\infty}\mathbf{X}_{-2}(\mathfrak{o}_{F_{\mathfrak{l}}})$. By Theorem 0.3, we only need to show that there exists $\beta \in \prod_{\mathfrak{l}\leqslant\infty}\mathbf{X}_{(-2)}(\mathfrak{o}_{F_{\mathfrak{l}}})$ such that $\psi_{\Theta/E}(f_E(\beta)) = 1$. By Lemma 3.1, we can choose $(a_{\mathfrak{p}}, b_{\mathfrak{p}}) \in \mathbf{X}_{-2}(\mathfrak{o}_{F_{\mathfrak{p}}})$ such that

$$\psi_{\Theta/E,v}(a_{\mathfrak{p}} + b_{\mathfrak{p}}\sqrt{-1}) = \left(\frac{2}{p}\right)_4 = 1.$$

Let $\beta = (\beta_{\mathfrak{l}})_{\mathfrak{l}\leqslant\infty} \in \prod_{l\leqslant\infty}\mathbf{X}_{-2}(\mathfrak{o}_{F_{\mathfrak{l}}})$ such that $\beta_{\mathfrak{p}} = (a_{\mathfrak{p}}, b_{\mathfrak{p}})$. Since Θ/E is unramified everywhere except at v, one obtains

$$\psi_{\Theta/E}(f_E(\beta)) = 1 \cdot \psi_{\Theta/E,v}(a_{\mathfrak{p}} + b_{\mathfrak{p}}\sqrt{-1}) = 1.$$

This completes the proof of sufficiency.

By Lemma 3.1, one has

$$\psi_{\Theta/E}\bigg(f_E\Big[\prod_{\mathfrak{p}\leqslant\infty}(x_{\mathfrak{p}}, y_{\mathfrak{p}})\Big]\bigg) = \left(\frac{2}{p}\right)_4 \text{ or } -1$$

for any $\prod_{\mathfrak{p}\leqslant\infty}(x_{\mathfrak{p}}, y_{\mathfrak{p}}) \in \prod_{p\leqslant\infty}\mathbf{X}_{(-2)}(\mathfrak{o}_{F_{\mathfrak{p}}})$. The necessity is obvious. □

Finally, we give more explanation about Theorem 0.3 by the following example.

Example 3.3. *Let $F = \mathbb{Q}(\sqrt{-34})$ and let $\alpha = 5+4\sqrt{-34}$. It is clear that $-\alpha$ is a sum of two integral squares in $\mathfrak{o}_F$ ($-\alpha = 5^2 + (2-\sqrt{-34})^2$). However, α is not a sum of two integral squares in $\mathfrak{o}_F$.*

Proof. We can see $34 = 5^2 + (-3)^2$. Then $r_1 = 1$ and $s_1 = 4$. And

$$(2-\sqrt{-1})^2 - (1+4\sqrt{-1}) = (4+\sqrt{-1})(1-\sqrt{-1})^2$$

Let $H_F = F(\sqrt{3-2\sqrt{-2}})$ be the Hilbert class field of F. Then

$$\Theta = E(\sqrt{2-\sqrt{-1}+\sqrt{1+4\sqrt{-1}}}) \quad \text{and} \quad H_E = E(\sqrt{1+4\sqrt{-1}}).H_F.$$

By some direct computations, we know that α satisfies the Artin condition of H_E and that α does not satisfy the Artin condition of Θ. Therefore the equation $x^2 + y^2 = \alpha$ is not solvable over $\mathfrak{o}_F$. □

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References

1 Colliot-Thélène J L, Xu F. Brauer-Manin obstruction for integral points of homogeneous spaces and representations by integral quadratic forms. Compos Math, 2009, 145: 309–363
2 Dirichlet G L. Einige neue Sätze über unbestimmte Gleichungen. Kgl Preuss Akad d Wissensch, 1920, 1: 221–236
3 Epstein P. Zur Auflösbarkeit der Gleichung $x^2 - Dy^2 = -1$. J Reine Angew Math, 1934, 171: 243–252
4 Harari D. Le défaut d'approximation forte pour les groups algébriques commutatifs. Algebra Number Theory, 2008, 2: 595–611
5 Ji C, Wang Y, Xu F. Sums of three squares over imaginary quadratic fields. Forum Math, 2006, 18: 585–601

6 Milne J S. Algebraic Geometry. Singapore: World Scientific, 1998
7 Nagell T. On the representations of integers as the sum of two integral squares in algebraic, mainly quadratic fields. Nova Acta Soc Sci Upsal (4), 1953, 15: 77pp
8 Nagell T. On the sum of two integral squares in certain quadratic fields. Ark Mat, 1961, 4: 267–286
9 Neukirch J, Schmidt A, Wingberg K. Cohomology of Number Fields. Grundlehren, vol. 323. Berlin: Springer, 2000
10 Niven I. Integers of quadratic fields as sums of squares. Trans Amer Math Soc, 1940, 48: 405–417
11 Qin H. The sum of two squares in a quadratic field. Comm Algebra, 1997, 25: 177–184
12 Rédei L. Über die Pellsche Gleichung $t^2 - du^2 = -1$. J Reine Angew Math, 1935, 173: 193–221
13 Siegel C L. Sums of mth powers of algebraic integers. Ann Math, 1935, 46: 313–339
14 Wei D. On the sum of two integral squares in quadratic fields $\mathbb{Q}(\sqrt{\pm p})$. Acta Arith, 2011, 147: 253–260
15 Wei D. On the diophantine equation $x^2 - Dy^2 = n$. Sci China Math, 2013, 56: 227–238
16 Wei D. On the sum of two integral squares in certain quadratic fields. Forum Math, doi: 10.1515/forum-2012-01677, in press
17 Wei D, Xu F. Integral points for multi-norm tori. Proc London Math Soc, 2012, 104: 1019–1044
18 Wei D, Xu F. Integral points for groups of multiplicative type. Adv Math, 2013, 232: 36–56

In：Bass H, Kuku A O, Pedrini C, eds. Algebraic *K*-Theory and Its Application. Singapore: World Scientific, 1999, 600－607

● Conference proceedings
Dmtriev V. Complete tables of the second rank constitutive tensors for linear homogeneous bianisotropic media described by point magnetic groups of symmetry and some general properties of the media. In: Proceedings of IEEE MTT-S IMOC'99. Berlin: Springer, 2000, 435－439

● Dissertation
Liu G X. Classification of finite dimensional basic Hopf algebras and related topics. Dissertation for the Doctoral Degree. Hangzhou: Hangzhou University, 2005, 24－28

● Electronic version of a journal
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